Brandon Matias

AI & Robotics Specialist

Bridging the gap between municipal infrastructure and autonomous intelligence. Specialized in applying Natural Language Processing and Computer Vision to complex urban and industrial challenges.

NLP & Financial Intelligence

NewsBot 2.0 (NLP)

Developed an automated news aggregation and sentiment classification engine. Mastered text cleaning, lexicon-based analysis, and trend visualization.

Fraud Detection Study

Analyzed Danske Bank's deep learning transition. Reduced false positives by 60% and increased detection rates by 50% through hidden pattern recognition.

Key Tools Mastered

Python, Pandas, NLTK, Sentiment Intensity Analyzers, Latent Feature Mapping.

CV & Industrial Automation

RE.Ai Solution (CV)

Implemented Computer Vision logic for automated virtual staging in real estate. Utilized image segmentation to identify spatial boundaries and assets.

GM Zero Downtime (ZDT)

Studied IIoT mesh networks at General Motors. Leveraged machine learning to detect "mechanical signatures" of failure before they occurred.

Mastery: Convolutional Neural Networks (CNNs), OpenCV, IoT Telemetry, Anomaly Detection.

Impact & Future Applications

Smart Cities

Deploying V2X communication and "Digital Moses" protocols to clear emergency corridors autonomously.

Healthcare

Implementing Explainable AI (XAI) in diagnostic imaging to ensure clinical transparency and patient safety.

Governance

Utilizing AGI-driven "Civic Mind" frameworks for equitable municipal resource management.

Wrap-up & Credentials

Applying rigorous data logic and ethical reasoning to solve the infrastructure problems of tomorrow.

View Technical Portfolio

https://github.com/BrandonLee-98/AI-Portfolio-Brandon-Matias

Contact Brandon

W216345386@student.hccs.edu

Image Sources

https://png.pngtree.com/thumb_back/fw800/background/20231006/pngtree-robotic-hand-enhancing-technology-connection-a-3d-conceptual-rendering-image_13570526.png

Source: pngtree.com

---

https://cdn.prod.website-files.com/653987c1e50cf8f938ffd18a/678554b6c889bb93e888b6b3_66575c6c312f54d43d65f470_commerical-first-person-fraud-01-header-image.webp

Source: www.experoinc.com

---

https://xmpro.com/wp-content/uploads/2024/01/Predictive-Maintenance-for-Robotic-Arms-in-the-Automotive-Industry-1536x650.jpg

Source: xmpro.com

---

https://www.mdpi.com/sustainability/sustainability-17-03254/article_deploy/html/images/sustainability-17-03254-g002.png

Source: www.mdpi.com